C12 Alkanes – Knowledge Organiser

AQA A-level Chemistry • Set 1: Crude Oil and Cracking **Paper 2**

#	Question	Answer
1	Crude oil	Mixture of mostly alkanes
2	Method used to separate crude oil	Fractional distillation
3	Property fractions are separated by	Boiling point
4	Hottest part of the fractionating column	Bottom
5	Where the shortest chains are collected	Top
6	Cracking	Breaking long-chain alkanes into shorter hydrocarbons
7	Why cracking is carried out	Shorter chains are more in demand
8	Conditions for thermal cracking	High temperature (up to 1000 °C), high pressure (70 atm)
9	Main products of thermal cracking	Alkenes
10	Conditions for catalytic cracking	Zeolite catalyst, about 450 °C, slight pressure

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AQA A-level Chemistry • Set 2: Combustion and Pollution **Paper 2**

#	Question	Answer
1	Main products of catalytic cracking	Branched alkanes, cycloalkanes, aromatics
2	Products of complete combustion	$CO_2 + H_2O$
3	Products of incomplete combustion	CO and/or C, + H_2O
4	Why CO is toxic	Binds to haemoglobin
5	Pollutant formed from N_2 in engines	NO_x
6	Pollutant formed from sulfur impurities	SO_2
7	Problem caused by SO_2	Acid rain
8	Substance used to remove SO_2 from flue gases	CaO or $CaCO_3$
9	Metals in a catalytic converter	Pt, Pd, Rh
10	Catalytic converter equation for CO and NO	$2CO + 2NO \rightarrow 2CO_2 + N_2$

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AQA A-level Chemistry • Set 3: Chlorination of Alkanes **Paper 2**

#	Question	Answer
1	Why alkanes are unreactive	Strong, non-polar C–C and C–H bonds
2	Three greenhouse gases	CO_2, CH_4, H_2O
3	Condition for chlorination of methane	UV light
4	Type of reaction of methane with chlorine	Free-radical substitution
5	Radical	Species with an unpaired electron
6	Three stages of a free-radical reaction	Initiation, propagation, termination
7	Initiation equation for chlorine	$Cl_2 \rightarrow 2Cl\bullet$
8	Products of methane with chlorine (one substitution)	$CH_3Cl + HCl$
9	Why a mixture of products forms	Further substitution
10	Stage where two radicals combine	Termination